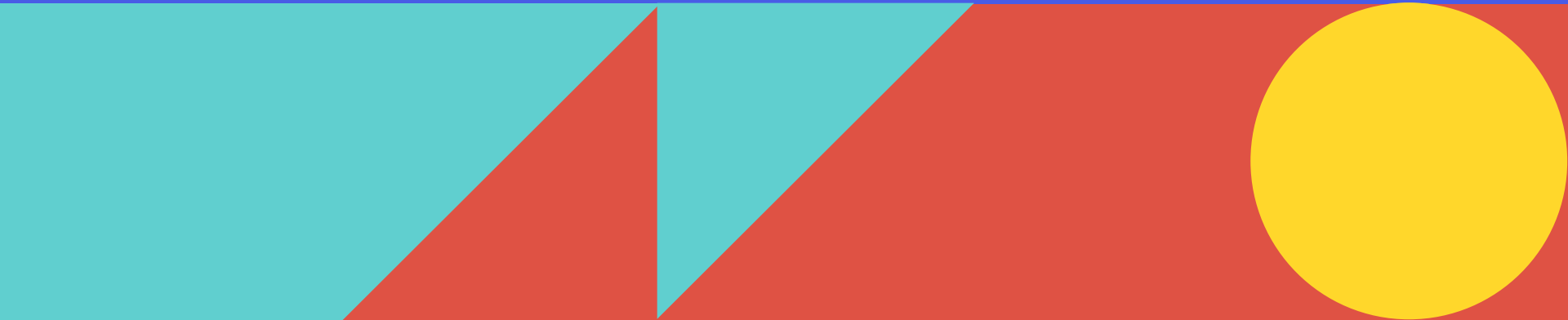


IP Crowdsourcing



Goals/ Pain Points

“What makes it hard to have meaningful conversations online?”

- Hard to find meaningful comments in the threads
- Hard to understand what people are communicating
- Opinions tend to be polarized/ emotional
- People may not be motivated to do the work of meaningful conversation



Things we Explored

Discouraging repetition

Avoiding repetition

Quantifying Intelligibility

Sentiment Analysis

UI/UX and Gamification

PNAMBIC

Pay No Attention to the Man Behind the Curtain

Implementations are not complete, but serve as a foundation for brainstorming and future work

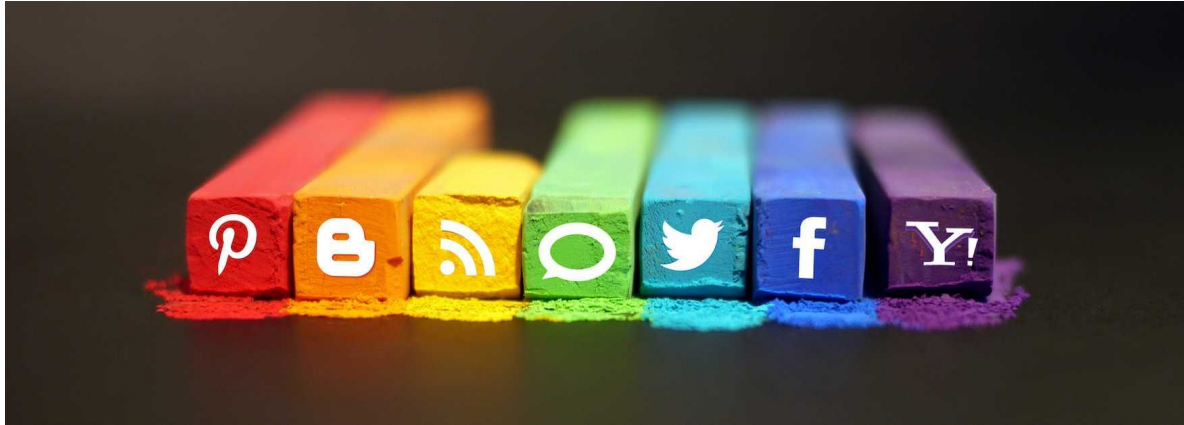
Some features are closer to being done than others

Discourage Repetition

Eric, Cole, and Kun-lin

Overall Goal

- Encourage users to make unique comments
- Utilize visualization to reduce repetition
- Reduce bias within posts



Design 1

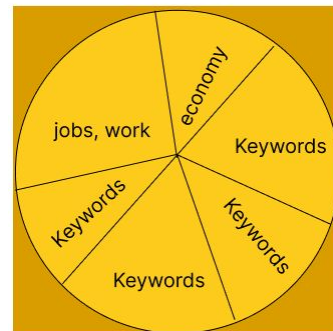
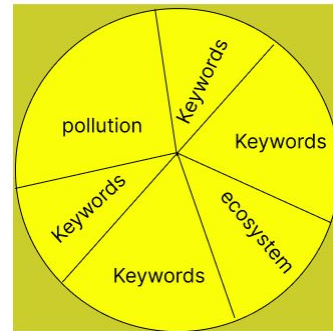
- Opinions shown in each column
- Color choice: Orange and Yellow
 - neutral colors: avoid bias
 - color blind-friendly
- Neutral Names
 - X and Y instead of agree disagree, or opinion 1 opinion 2
- Visualizer
 - Helps users easily see various topics

Post

We should stop the construction of this factory!

Comments

Opinion X	Opinion Y
Agree, pollution is bad	The factory is needed for jobs.
Disrupt ecosystems in the area	The economy will greatly benefit.
comment	comment
Pollution will dirty the environment	I need a place to work.
comment	comment
comment	comment
comment	comment



Design 2

- Color choice: Blue and Orange
 - Blue means it is more liked and orange is less liked
 - Color only shows after you interact (as in the iron man option)
- A way to agree or disagree with a comment
- The option to respond to and express opinion on each individual comment
 - By clicking the arrow on the right
- Displays both liked and disliked content

Who is your favorite superhero?

Hawkeye

Agree

Disagree



Iron Man

Agree

Disagree



Captain America

Agree

Disagree



Dr. Evil

Agree

Disagree



The Green Lantern

Agree

Disagree



Clustering Comments

Why cluster and rank comments?

- The need to summarize user-contributed comments to extract key information and the problems with existing methods to summarize information.

1. Human Bias:

As is the case with editorials the humans select comments to represent a range of comments that represent a range of views. Biased towards the world view of the comment selector.

2. Collaborative recommendation:

Allows user to themselves represent most valuable comments. Aggregates community perspective but does not reflect informative comments or favors comments by popular users.

3. Keyword Cloud:

Represent a wide range of comments that may not contain repetitions but lack context.

Cosine Similarity

1. Vectorize phrases it using Glove(Matrix) or Word2Vec(Neural Network)
 - a. Uses TF:IDF to get vector

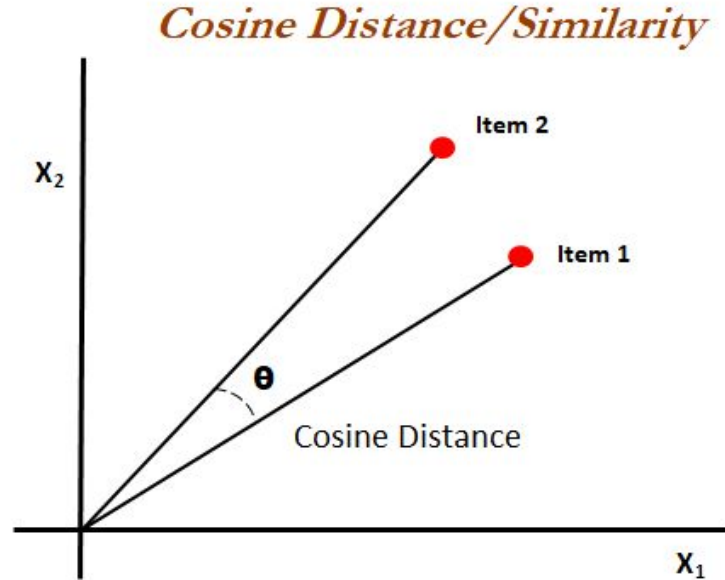
$$w_{i,j} = tf_{i,j} \times \log\left(\frac{N}{df_i}\right)$$

tf_{ij} = number of occurrences of i in j

df_i = number of documents containing i

N = total number of documents

2. Creates a vector map where similar sentences have smaller angles



K Means Clustering

Advantages

- Can quickly large data sets
- Always converges

Disadvantages

- Optimal cluster number is must be found
- Curse of dimensionality

Comment	Votes	Time	Tokenized Comment	cluster_label
wow dadsometimes the hard decision is the right decision stay strong	1	1625343221.0	['wow', 'dadsometim', 'hard', 'decis', 'right', 'decis', 'stay', 'strong']	1
nta stay strong bro you are completely in the right on this one just because they are cousins does not mean they are entitled to your daughters belongings if they wanted that makeup so bad they should go buy their own and as you said this is probably one of the many reasons outside of your wife why his own family would not take them in	1	1625339378.0	['nta', 'stay', 'strong', 'bro', 'complet', 'right', 'one', 'cousin', 'mean', 'entitl', 'daughter', 'belong', 'want', 'makeup', 'bad', 'go', 'buy', 'said', 'probabl', 'one', 'mani', 'reason', 'outsid', 'wife', 'famili', 'would', 'take']	1
nta you are the <u>only one</u> doing something that makes sense and taking action to help your daughter and people even family coming and living in someone elses home should have more respect	1	1625344595.0	['nta', 'one', 'someth', 'make', 'sens', 'take', 'action', 'help', 'daughter', 'peopl', 'even', 'famili', 'come', 'live', 'someon', 'els', 'home', 'respect']	1
nta not even one tiny bit an individual's propertybelongings should be respected also borrowing would imply someone had permission to take something if no one asked before doing so that is stealing stay strong	1	1625338850.0	['nta', 'even', 'one', 'tini', 'bit', 'individu', 's', 'propertybelong', 'respect', 'also', 'borrow', 'would', 'impli', 'someon', 'permiss', 'take', 'someth', 'one', 'ask', 'steal', 'stay', 'strong']	1
nta and you are amazing for sticking up for your daughter your bil and his children need a wake up call you are the one allowing them to stay in your home you would think that they would be a little more grateful but i guess that is how strong their entitlement is very sad and even worse on your wife's part	1	1625325515.0	['nta', 'amaz', 'stick', 'daughter', 'bil', 'children', 'need', 'wake', 'call', 'one', 'allow', 'stay', 'home', 'would', 'think', 'would', 'littl', 'grate', 'guess', 'strong', 'entitl', 'sad', 'even', 'wors', 'wife', 's', 'part']	1
borrowing is something one does with permission this is thieving nta you are the <u>only one</u> in that house who has your daughters back	1	1625336869.0	['borrow', 'someth', 'one', 'permiss', 'thiev', 'nta', 'one', 'hous', 'daughter', 'back']	1
nta stay strong you are obviously protecting your daughter and her right to privacy taking things without asking is not sharing this is stealing	1	1625343794.0	['nta', 'stay', 'strong', 'obvious', 'protect', 'daughter', 'right', 'privaci', 'take', 'thing', 'without', 'ask', 'share', 'steal']	1
do not get marry and have kids just spread your seed and hope that one becomes strong enough to surpass you	0	1625348605.0	['get', 'marri', 'kid', 'spread', 'seed', 'hope', 'one', 'becom', 'strong', 'enough', 'surpass']	1
5 you are not the a your wife is the a for this one	1	1625339798.0	['wife', 'one']	1

How Many Clusters?

- Elbow Method is not objective and has high runtime
 - For 5800 comments testing from 1-100 clusters took around 300 seconds
- Silhouette Score is objective, but has high runtime
 - For 5800 comments testing from 1-100 clusters took around 400 seconds

What next?

- Order within cluster
- Faster Cluster Number Calculation

Quantifying Intelligibility

David, Matthew, Vanshika, Rohit, Veronica

What is Intelligibility?

Hamburg Comprehensibility Concept

- Simplicity (difficulty and understandability)
- Structure (logical)
- Conciseness (to the point and clear)

Discourse Relations: textual structures that produce discussion

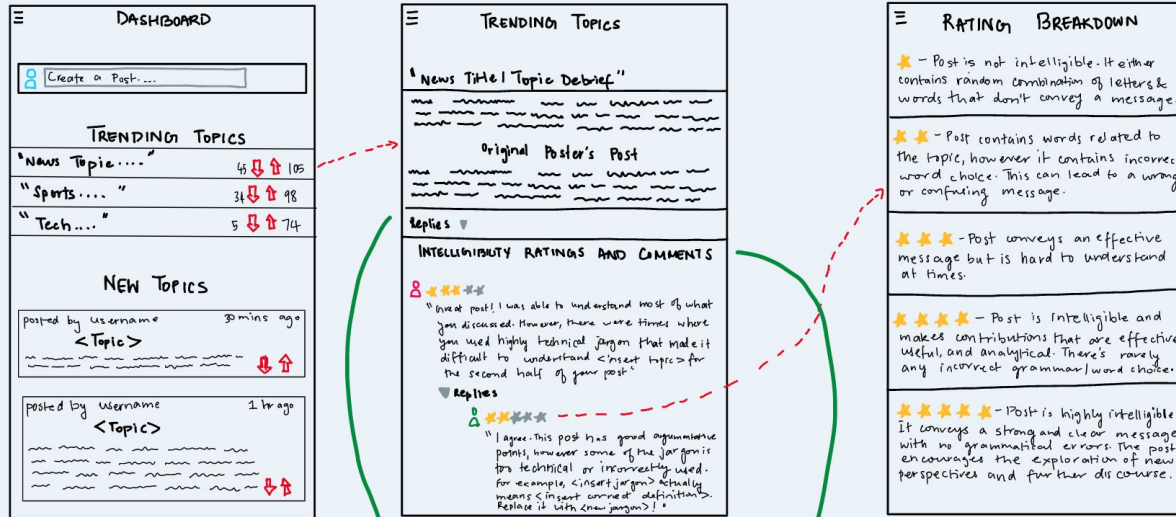
Relevancy: categorization algorithm

Supervised Model

- Users rate comments and posts
- Collect sufficient data from existing posts for analysis
- Train a supervised model based on quantifiable attributes (comment length, filler words, likes, etc.)
- Predict the intelligibility scores of future comments



UI Prototype



Traditional comment section much like those on platforms such as Reddit, Youtube, Twitter, etc

VS

Specialized comment section for user's ratings, opinions and other comments specific to the intelligibility of this post.

Feature Implemented: Feedback (Commenting & Rating) System

Purpose 1

Allows users to leave comments about a post's intelligibility (difficulty and understanding of the topic). Comments can include feedback, suggestions or specific corrections/edits to the original post.

Purpose 2

Allows the users to leave a 5 star scale rating that characterizes the post as specified in the breakdown. Users can access the breakdown by clicking on a rating another user has given.

The system is META

- A system that supplies tools to supervise and manage another system
- Implements a feedback system that ensures quality and intelligibility of the main system

Intelligibility Algorithm

- Use text similarity nlp libraries to vectorize sentences in a post
 - Compare comments to the original post to check if the topic stays the same.
- Vectors are scaled from 0 - 1
 - 0 represents the posts are entirely different
 - 1 identifies the sentences as the exact same
 - Regular range goes from ~.3 - .8
 - Scores greater than .8 may signify repetition
- Ideally check to see if posts are in the range
 - Outliers may just mean the conversation is not topical

```
The raters start map ablations post-MRIs ffor patients RA ITK-snap.  
Once ablations mapped analyzed overlaid FreeSurfer 3D model patient's brain order differences raters determine huge variances analyses.  
Afterward, ablations analyzed determine parts o white gray matter cover determine parts brain ablated led success procedure.  
#raters #ablations #post #mris #ffor #patients #ra #snap  
#ablations #freesurfer #3d #model #patient #brain #order #differences #raters #huge #variances #analyses  
#ablations #parts #o #white #gray #matter #parts #brain #success #procedure  
[0.9184054915175073, 0.9046042632748642]
```

Summary

Overall, our categorization algorithm design works by:

1. defining intelligibility similarities
2. Collecting user ratings
3. Applying ratings to model
4. Quantifying post

This runs under the UI features

The goal is to train a model to predict the intelligibility of comments through aggregating large amounts of user rankings.

In order to achieve this:

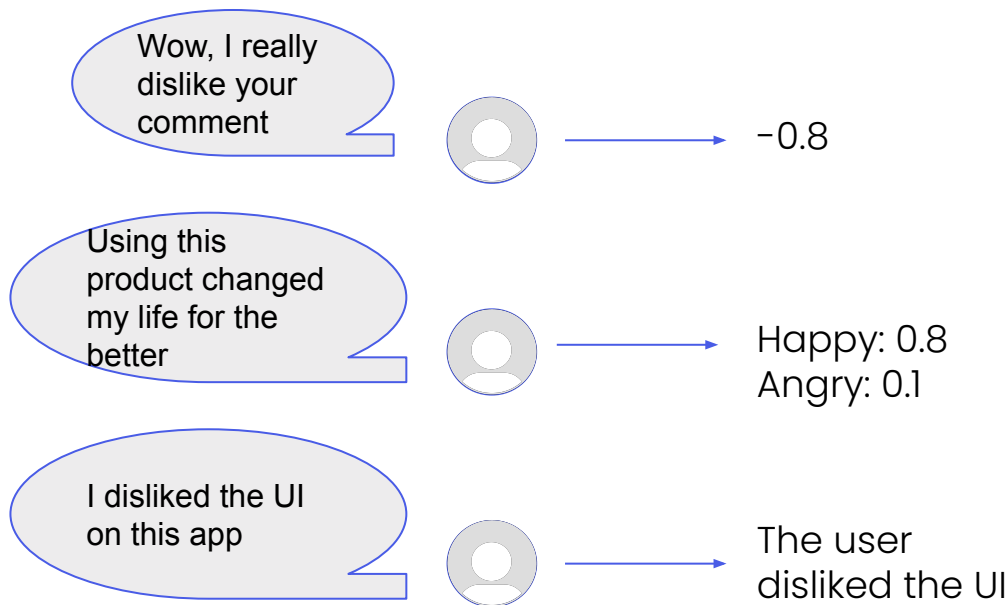
We require data to train but we cannot have repetitive data, so we have a similarity algorithm to weed out repetitions/duplicate posts

Sentiment Analysis

Ian BaracsKay David Liu, Matthew Jimenez,
Willa Zheng,

What is Sentiment Analysis?

- Positive/Negative model
- Emotion detection model
- Aspect-based model

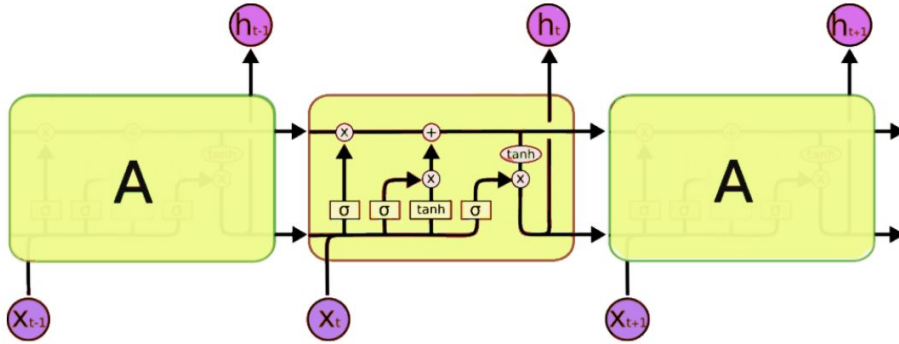


Goal and Guiding Questions:

Goal: How can we manage the diverse range of people's opinions and emotions in a conversation?

- Should we show the results of our sentiment analysis algorithm to the user?
 - How should this be done?
 - How much information should we reveal?
- How should the results of our sentiment analysis influence the exposure of comments on our social media?
 - Should we be "ascending" or promoting comments that are more positive/neutral?

LSTM Sentiment Analysis Model

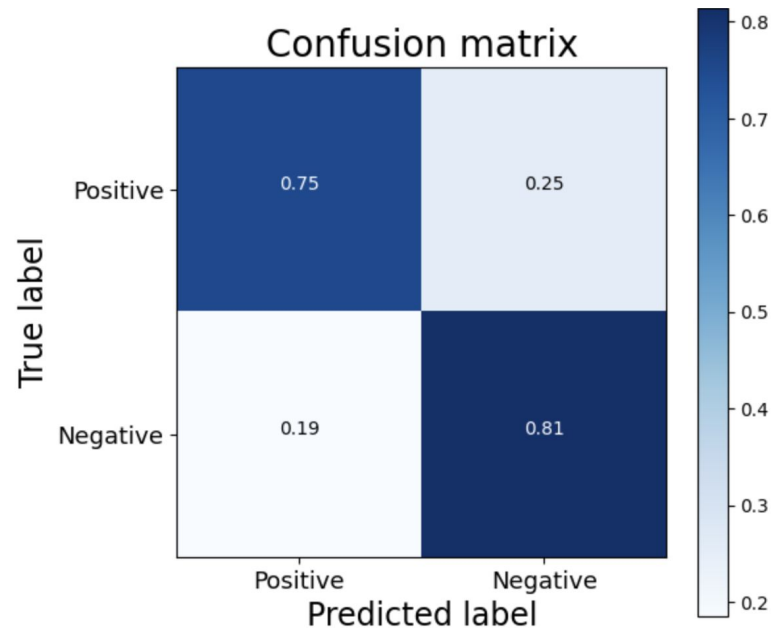
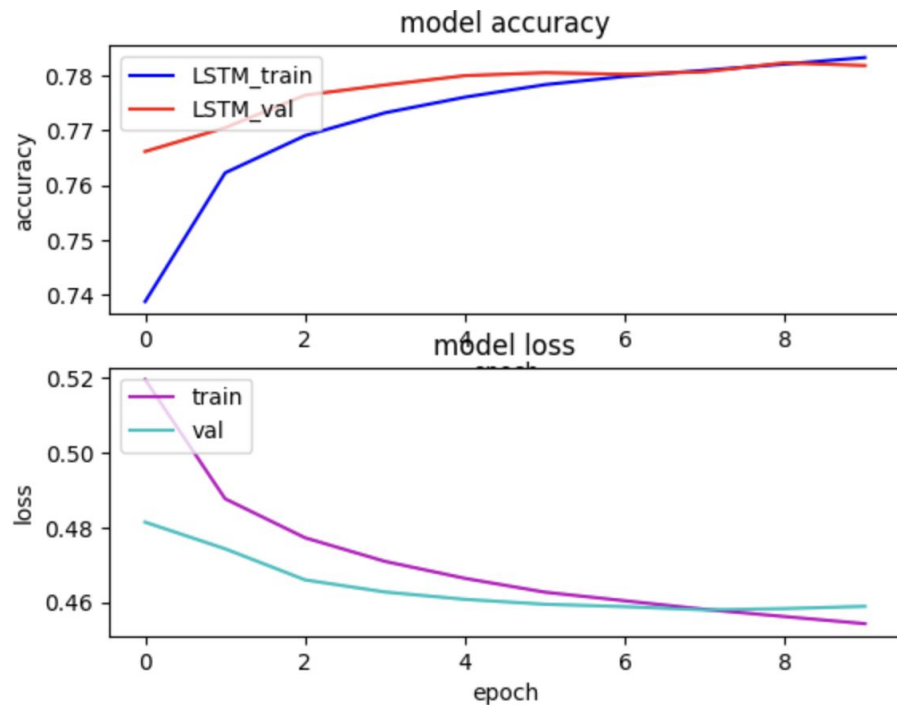


- Feedback connections
- Vanishing gradient problem

Future Plans:

- Emotion detection model

Results?



Some Options for comments

Default

	@VeryAngryMan	Responding To: @verystupidman
Oh my God! I am so angry right now. I can't believe someone on the internet would have such a dumb opinion!!!		

Showing only +/-

	@VeryAngryMan	Responding To: @verystupidman
Oh my God! I am so angry right now. I can't believe someone on the internet would have such a dumb opinion!!!		

Show Everything

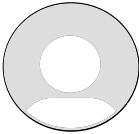
	@VeryAngryMan	Responding To: @verystupidman
Oh my God! I am so angry right now. I can't believe someone on the internet would have such a dumb opinion!!!		-0.89 

Indicates
negative
sentiment

Emotion-detection

Post/Threads Sentiment Analysis

- Average together the sentiments of the comments to get an idea of the general emotions of the post and its comments.
 - Gives a user an idea of what they're going into
 - Helps with self-moderation of discussion
- Should there be manipulation on comments to try and "influence" more positive conversations
 - Sometimes conversations should be sad or negative
 - It would be weird if the polar bear post had lots of happy or positive comments...
- Middle ground could be found through experimentation



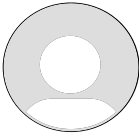
@NumberOneLethalWeaponFan

Oh my gosh, Lethal Weapon 7 just announced to come out 2024!!!



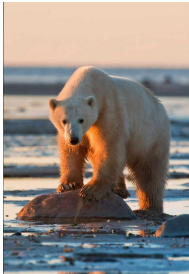
Emotions:

Happy - 0.82
Sad - 0.24
Angry - 0.12



@NumberOnePandaBearFan

Scientists predict polar bears will go extinct by 2030 (Source: scientists)



Emotions:

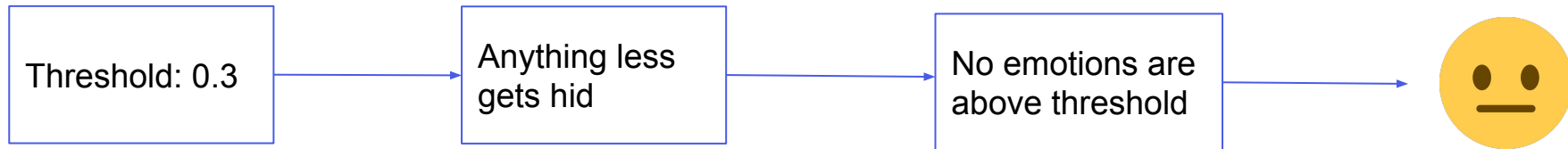
Sad - 0.8
Angry - 0.6
Disgust - 0.12

***Could also do this on comment threads

Uncertainty/Inaccuracies



- Neutral opinions will be hard to categorize emotions for.
- No model is ever perfect and users might get upset if their comment is labeled angry when it might not actually be.
 - Certainty Measurement?
 - If the model is uncertain on any emotion at all, you could not display anything at all or give a neutral face.



Outstanding Questions

- When should the sentiment analysis actually be run?
 - As the comment is being written
 - Allow users to send analysis requests
 - When the user attempts to post their comment
 - How much will this slow down the website as a whole?
- How would users react to seeing the sentiment of their own comments?
 - Urge to rewrite it?
 - Make them angry?
- Would users be influenced to post comments on posts with sentiments displayed?

Future Plans

- Get our very own working emotion-detection model.
 - Further in the future, an aspect-based model might be possible to design and attempt to implement
- Research/experiment how users will react to seeing an AI classify the sentiment of their comments for others to see.
 - Additionally, research how users would behave with sentiment information displayed on other comments/posts.
 - Exploits?

Guiding With UI/UX

Yoon Ji, Shree, Asim, Suhas, Chandreyi

What is the need for gamification?

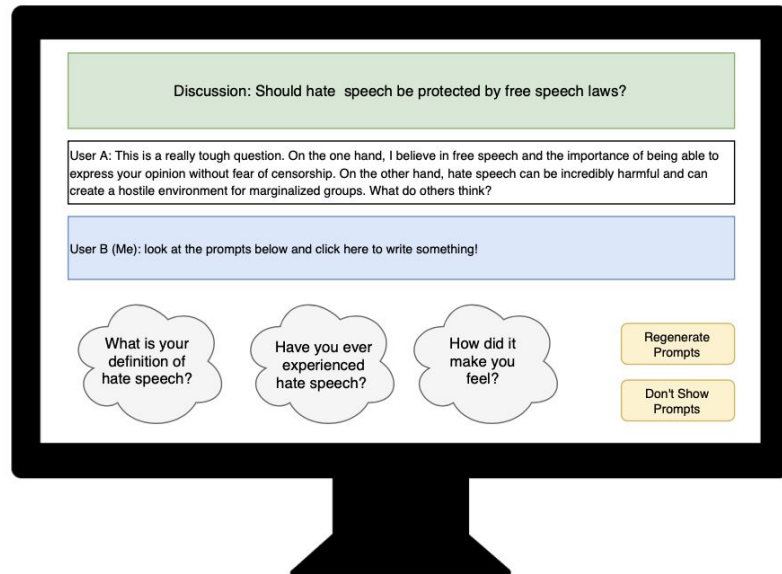
1) Increases engagement

2) Fosters sense of community

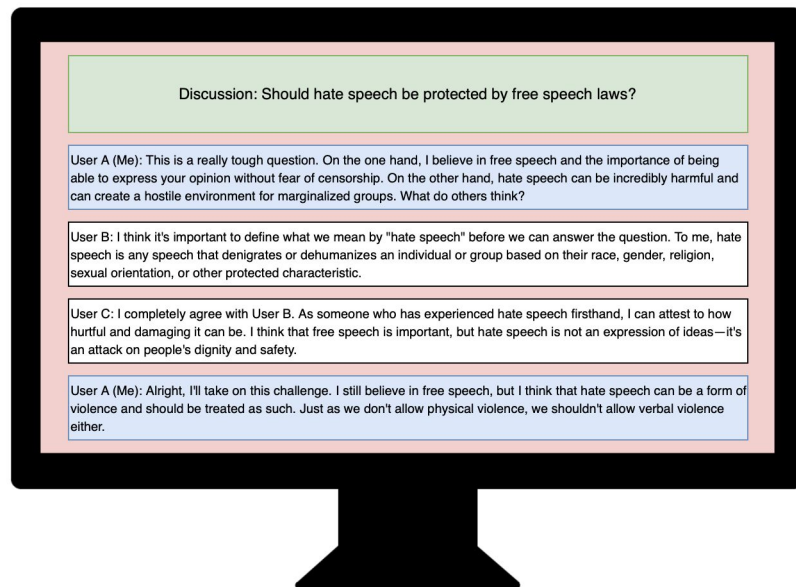
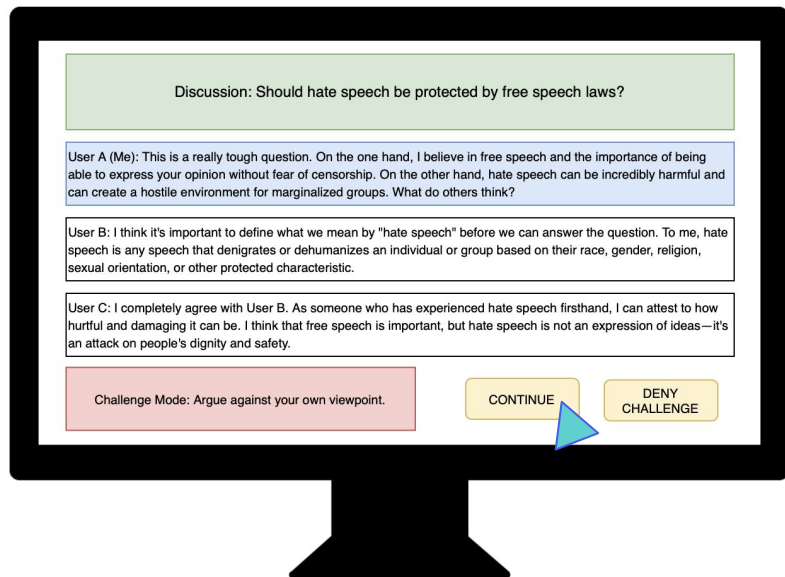
3) Grow size of user base



Prompts – Prompts Generation

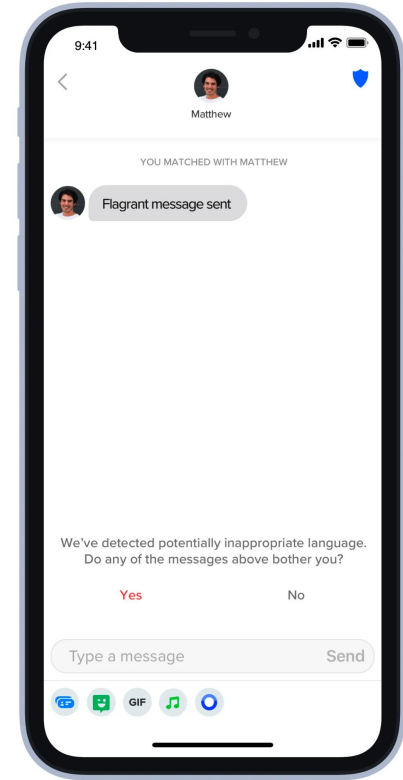


Prompts - Challenge Mode

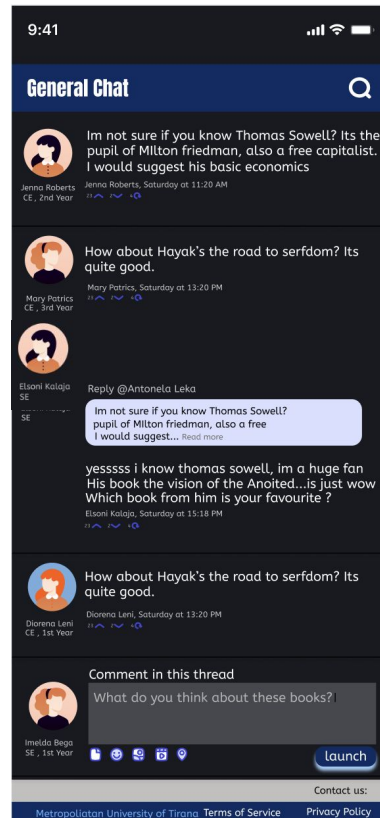
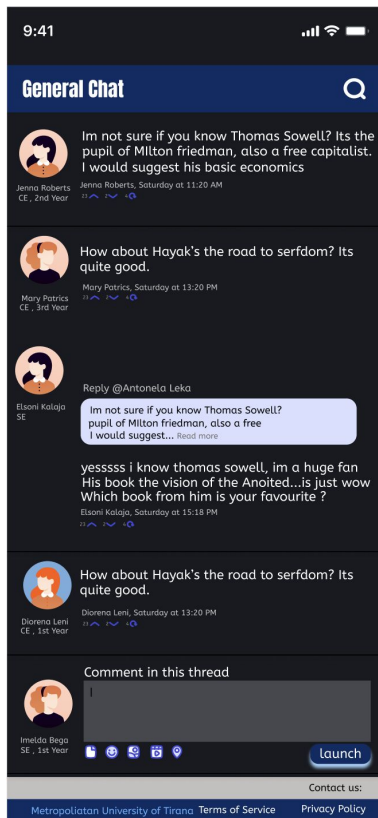


Pop-Ups

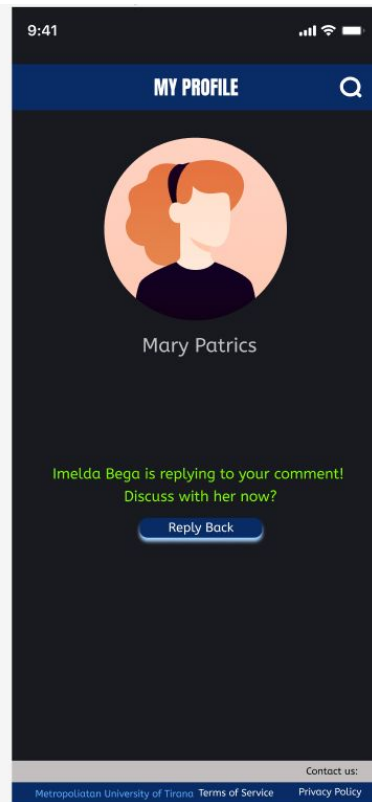
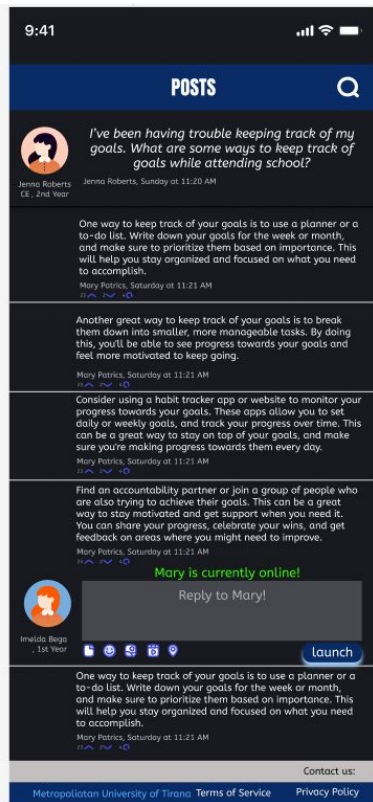
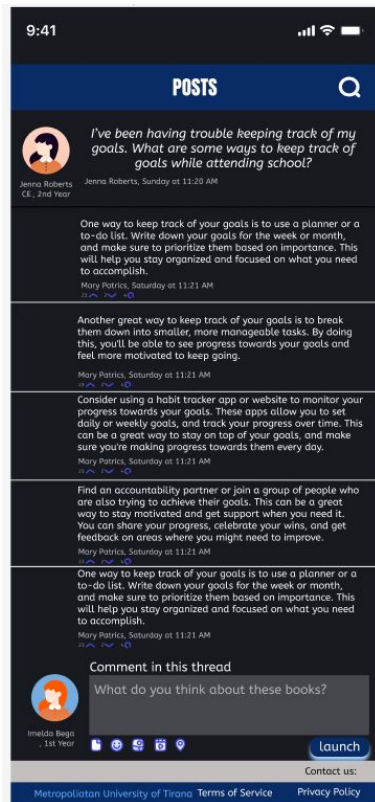
- Grabs user attention
- Inform users before comments are posted about inappropriate language
- Points notification system
- Generic notifications (“Cite your sources”)



Encouraging Relevant Discussion



Real-Time Discussions



Thank You!

